Formally Verified PCP Constructions

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Chapter 1

Linearity test over finite fields

1.1 Basic tools for the analysis of functions over finite fields

Throughout this section, let $q = p^s$ for a prime p and $s \in \mathbb{N}$ with $s \geq 1$. All vectors are in $\mathbb{F}_q^n$, and all expectations are uniform over the indicated finite sets. Let $\omega_p = e^{2\pi i/p}$.

Definition 1.1.1 (Field trace)**.** The field trace $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is

$$\mathrm{Tr}(z) := \sum_{i=0}^{s-1} z^{p^i}.$$

Definition 1.1.2 (Base additive character)**.** The base additive character $\psi : \mathbb{F}_q \to \mathbb{C}$ is

$$\psi(z) := \omega_p^{\mathrm{Tr}(z)}.$$

Definition 1.1.3 (Additive characters)**.** For $\alpha \in \mathbb{F}_q^n$, define the additive character $\chi_\alpha : \mathbb{F}_q^n \to \mathbb{C}$ by

$$\chi_\alpha(x) := \psi(\langle \alpha, x \rangle) = \omega_p^{\mathrm{Tr}(\langle \alpha, x \rangle)},$$

where $\langle \alpha, x \rangle = \sum_{i=1}^n \alpha_i x_i \in \mathbb{F}_q$.

Definition 1.1.4 (Expectation)**.** For a function $h : \mathbb{F}_q^n \to \mathbb{C}$, define its uniform expectation by

$$\mathbb{E}_{x \in \mathbb{F}_q^n}[h(x)] := \frac{1}{q^n} \sum_{x \in \mathbb{F}_q^n} h(x).$$

Definition 1.1.5 (Inner product)**.** For functions $h_1, h_2 : \mathbb{F}_q^n \to \mathbb{C}$, define

$$\langle h_1, h_2 \rangle := \mathbb{E}_{x \in \mathbb{F}_q^n} \left[h_1(x) \overline{h_2(x)} \right].$$

Definition 1.1.6 (L_2 norm)**.** For a function $h : \mathbb{F}_q^n \to \mathbb{C}$, define its L_2 norm by

$$\|h\|_2 := \left(\mathbb{E}_{x \in \mathbb{F}_q^n} \left[|h(x)|^2 \right] \right)^{1/2} = \sqrt{\langle h, h \rangle}.$$

Definition 1.1.7 (Fourier coefficient)**.** For $h : \mathbb{F}_q^n \to \mathbb{C}$ and $\alpha \in \mathbb{F}_q^n$, define

$$\hat{h}(\alpha) := \langle h, \chi_\alpha \rangle = \mathbb{E}_{x \in \mathbb{F}_q^n} \left[h(x) \overline{\chi_\alpha(x)} \right].$$

Lemma 1.1.8 (Characters are orthonormal)**.** *For every* $\alpha, \beta \in \mathbb{F}_q^n$*,*

$$\langle \chi_\alpha, \chi_\beta \rangle = \begin{cases} 1 & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta. \end{cases}$$

Proof. First observe that for every $\alpha, \beta \in \mathbb{F}_q^n$ and every $x \in \mathbb{F}_q^n$,

$$\chi_\alpha(x)\overline{\chi_\beta(x)} = \omega_p^{\mathrm{Tr}(\langle \alpha, x \rangle)} \omega_p^{-\mathrm{Tr}(\langle \beta, x \rangle)} = \omega_p^{\mathrm{Tr}(\langle \alpha - \beta, x \rangle)} = \chi_{\alpha-\beta}(x),$$

where we used the $\mathbb{F}_p$-linearity of the trace map.

If $\alpha = \beta$, then $\alpha - \beta = 0$, so $\chi_{\alpha-\beta}(x) = 1$ for every $x \in \mathbb{F}_q^n$. Hence

$$\mathbb{E}_{x \in \mathbb{F}_q^n} \left[\chi_\alpha(x)\overline{\chi_\beta(x)} \right] = \mathbb{E}_{x \in \mathbb{F}_q^n}[1] = 1.$$

Now suppose $\alpha \neq \beta$. Let $\gamma := \alpha - \beta$. Then $\gamma \neq 0$, so there exists an index j such that $\gamma_j \neq 0$. Fix all coordinates of x except x_j. For the fixed remaining coordinates, the map

$$x_j \mapsto \langle \gamma, x \rangle$$

has the form

$$x_j \mapsto \gamma_j x_j + c$$

for some constant $c \in \mathbb{F}_q$. Since $\gamma_j \neq 0$, multiplication by γ_j is a bijection on $\mathbb{F}_q$, and therefore the affine map $x_j \mapsto \gamma_j x_j + c$ is also a bijection on $\mathbb{F}_q$. Thus, as x_j ranges uniformly over $\mathbb{F}_q$, the value $\langle \gamma, x \rangle$ also ranges uniformly over $\mathbb{F}_q$. Averaging first over x_j and then over the remaining coordinates gives

$$\mathbb{E}_{x \in \mathbb{F}_q^n} \left[\chi_\gamma(x) \right] = \mathbb{E}_{t \in \mathbb{F}_q} \left[\omega_p^{\mathrm{Tr}(t)} \right].$$

It remains to show that this last average is zero. The trace map $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is a nonzero $\mathbb{F}_p$-linear map, so every fiber of Tr has the same cardinality. Since $|\mathbb{F}_q| = q$ and $|\mathbb{F}_p| = p$, each fiber has cardinality q/p. Therefore

$$\begin{aligned} \mathbb{E}_{t \in \mathbb{F}_q} \left[\omega_p^{\mathrm{Tr}(t)} \right] &= \frac{1}{q} \sum_{t \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(t)} \\ &= \frac{1}{q} \sum_{r \in \mathbb{F}_p} \sum_{\substack{t \in \mathbb{F}_q \\ \mathrm{Tr}(t) = r}} \omega_p^r \\ &= \frac{1}{q} \sum_{r \in \mathbb{F}_p} \frac{q}{p} \omega_p^r \\ &= \frac{1}{p} \sum_{r \in \mathbb{F}_p} \omega_p^r \\ &= 0, \end{aligned}$$

because the sum of all p-th roots of unity is zero. Hence, if $\alpha \neq \beta$, then

$$\mathbb{E}_{x \in \mathbb{F}_q^n} \left[\chi_\alpha(x)\overline{\chi_\beta(x)} \right] = 0.$$

Combining the two cases proves the stated orthogonality relation. The equivalent formulation follows by taking $\beta = 0$. $\square$

Lemma 1.1.9 (Fourier inversion)**.** *For every function $h : \mathbb{F}_q^n \to \mathbb{C}$ we have*

$$h(x) = \sum_{\alpha \in \mathbb{F}_q^n} \widehat{h}(\alpha)\chi_\alpha(x) \quad \text{for every } x \in \mathbb{F}_q^n.$$

Proof. By Theorem 1.1.8, the additive characters

$$\{\chi_\alpha : \alpha \in \mathbb{F}_q^n\}$$

are orthonormal with respect to the inner product

$$\langle h_1, h_2 \rangle = \mathbb{E}_{x \in \mathbb{F}_q^n}\left[h_1(x)\overline{h_2(x)}\right].$$

Moreover, these characters span the vector space of all complex-valued functions on $\mathbb{F}_q^n$. Indeed, there are exactly q^n characters, and the space of functions $\mathbb{F}_q^n \to \mathbb{C}$ has dimension q^n. Since the characters are nonzero and pairwise orthogonal, they are linearly independent. Hence they form an orthonormal basis.

Therefore every function $h : \mathbb{F}_q^n \to \mathbb{C}$ has a unique expansion in this basis:

$$h = \sum_{\alpha \in \mathbb{F}_q^n} c_\alpha \chi_\alpha$$

for some coefficients $c_\alpha \in \mathbb{C}$. Taking the inner product of both sides with χ_β, we obtain

$$\langle h, \chi_\beta \rangle = \sum_{\alpha \in \mathbb{F}_q^n} c_\alpha \langle \chi_\alpha, \chi_\beta \rangle.$$

By orthonormality, all terms in the sum vanish except the term $\alpha = \beta$, and $\langle \chi_\beta, \chi_\beta \rangle = 1$. Hence

$$\langle h, \chi_\beta \rangle = c_\beta.$$

By definition of the Fourier coefficient,

$$\widehat{h}(\beta) = \langle h, \chi_\beta \rangle.$$

Thus $c_\beta = \widehat{h}(\beta)$ for every $\beta \in \mathbb{F}_q^n$. Therefore

$$h = \sum_{\alpha \in \mathbb{F}_q^n} \widehat{h}(\alpha)\chi_\alpha.$$

Evaluating this identity at $x \in \mathbb{F}_q^n$ gives the Fourier inversion formula

$$h(x) = \sum_{\alpha \in \mathbb{F}_q^n} \widehat{h}(\alpha)\chi_\alpha(x).$$

□

Lemma 1.1.10 (Plancherel identity)**.** *For every pair of functions $f, g : \mathbb{F}_q^n \to \mathbb{C}$,*

$$\sum_{\alpha \in \mathbb{F}_q^n} \widehat{f}(\alpha)\overline{\widehat{g}(\alpha)} = \langle f, g \rangle.$$

Proof. By Theorem 1.1.9,

$$f = \sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\chi_\alpha.$$

Taking the inner product with g and using linearity in the first argument gives

$$\langle f, g\rangle = \sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\langle \chi_\alpha, g\rangle.$$

Since

$$\langle \chi_\alpha, g\rangle = \overline{\langle g, \chi_\alpha\rangle} = \overline{\hat{g}(\alpha)},$$

we obtain

$$\langle f, g\rangle = \sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\overline{\hat{g}(\alpha)}.$$

□

Lemma 1.1.11 (Parseval identity)**.** *For every function $h : \mathbb{F}_q^n \to \mathbb{C}$,*

$$\sum_{\alpha \in \mathbb{F}_q^n} \left|\hat{h}(\alpha)\right|^2 = \|h\|_2^2.$$

Proof. Apply Theorem 1.1.10 with $f = g = h$:

$$\sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha)\overline{\hat{h}(\alpha)} = \langle h, h\rangle.$$

The left-hand side is

$$\sum_{\alpha \in \mathbb{F}_q^n} \left|\hat{h}(\alpha)\right|^2,$$

and by the definitions of the inner product and the L_2 norm, the right-hand side is

$$\langle h, h\rangle = \mathbb{E}_{x \in \mathbb{F}_q^n}\left[|h(x)|^2\right] = \|h\|_2^2.$$

□

Definition 1.1.12 (Nonzero field elements)**.** Write

$$\mathbb{F}_q^\times := \{c \in \mathbb{F}_q : c \neq 0\}.$$

Definition 1.1.13 (Fourier expansion of a finite-field-valued function)**.** Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$. For each $c \in \mathbb{F}_q^\times$, define the c-phase lift

$$\varphi_c(x) := \omega_p^{\mathrm{Tr}(cf(x))}.$$

The Fourier expansion of f is the set $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$.

Lemma 1.1.14 (Character-sum indicator)**.** *For every $u, v \in \mathbb{F}_q$,*

$$\mathbb{1}[u = v] = \frac{1}{q}\sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(u-v))}.$$

Proof. For each $z \in \mathbb{F}_q$ we define

$$\psi(z) := \omega_p^{\mathrm{Tr}(z)}$$

be the standard additive character of $\mathbb{F}_q$. We prove that, for every $t \in \mathbb{F}_q$,

$$\mathbb{1}[t=0] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct).$$

The desired statement follows by taking $t = u - v$.

If $t = 0$, then $ct = 0$ for every $c \in \mathbb{F}_q$, so

$$\frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct) = \frac{1}{q} \sum_{c \in \mathbb{F}_q} 1 = 1.$$

Now suppose $t \neq 0$. Then multiplication by t is a bijection $\mathbb{F}_q \to \mathbb{F}_q$, so

$$\sum_{c \in \mathbb{F}_q} \psi(ct) = \sum_{z \in \mathbb{F}_q} \psi(z).$$

We claim that the latter sum is 0. Let

$$S := \sum_{z \in \mathbb{F}_q} \psi(z).$$

Since $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is a nonzero $\mathbb{F}_p$-linear map, it is surjective. Hence there exists $a \in \mathbb{F}_q$ such that $\mathrm{Tr}(a) = 1$. Using the bijection $z \mapsto z + a$, we get

$$S = \sum_{z \in \mathbb{F}_q} \psi(z+a) = \sum_{z \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(z+a)} = \sum_{z \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(z)} \omega_p^{\mathrm{Tr}(a)} = \omega_p S.$$

Since $\omega_p \neq 1$, this implies $S = 0$. Therefore, when $t \neq 0$,

$$\frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct) = 0.$$

Combining the two cases gives

$$\mathbb{1}[t=0] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(ct)}.$$

Substituting $t = u - v$, we obtain

$$\mathbb{1}[u=v] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(u-v))}.$$

□

Definition 1.1.15 (Relative Hamming distance)**.** For $f, g : \mathbb{F}_q^n \to \mathbb{F}_q$, define

$$\delta(f,g) := \Pr_{x \in \mathbb{F}_q^n} \left[f(x) \neq g(x) \right].$$

1.2 Linear functions over finite fields

Definition 1.2.1 (Linear functions)**.** For $\alpha \in \mathbb{F}_q^n$, define the linear scalar function $\ell_\alpha : \mathbb{F}_q^n \to \mathbb{F}_q$ by

$$\ell_\alpha(x) := \langle \alpha, x \rangle.$$

A function $h : \mathbb{F}_q^n \to \mathbb{F}_q$ is linear if $h = \ell_\alpha$ for some $\alpha \in \mathbb{F}_q^n$. The finite set of all linear scalar functions is

$$\mathcal{LIN} := \left\{ \ell_\alpha \mid \alpha \in \mathbb{F}_q^n \right\}.$$

Definition 1.2.2 (Distance to linearity)**.** The distance from $f : \mathbb{F}_q^n \to \mathbb{F}_q$ to linearity is

$$\delta(f, \mathcal{LIN}) := \min_{g \in \mathcal{LIN}} \delta(f, g).$$

Lemma 1.2.3 (Well-separatedness of $\mathcal{LIN}$)**.** *For any distinct $f, g \in \mathcal{LIN}$, one has $\delta(f, g) = 1 - 1/|\mathbb{F}|$.*

Proof. Let $a, b \in \mathbb{F}^n$ be the unique vectors such that $f(x) = \langle a, x \rangle$ and $g(x) = \langle b, x \rangle$ for each x. Let $c = a - b$, which must be a non-zero vector. In particular, there exists $i \in [n]$ such that $c_i \neq 0$. Take any permutation $\sigma : [n] \to [n]$ such that $\sigma(i) = n$. Then

$$\begin{aligned}
\delta(f, g) &= 1 - \Pr_{x \sim \mathbb{F}^n}[\langle c, x \rangle = 0] \\
&= 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} \sum_{x_n \in \mathbb{F}} \mathbb{1}[c_i \cdot x_n = - \sum_{j \in [n]\ i} c_j \cdot x_{\sigma(j)}] \\
&= 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} \sum_{x_n \in \mathbb{F}} \mathbb{1}[x_n = -c_i^{-1} \cdot \sum_{j \in [n]\ i} c_j \cdot x_{\sigma(j)}]
\end{aligned}$$

where the last equality follows because $c_i \neq 0$. Now note that $\mathbb{1}[x_n = -c_i^{-1} \cdot \sum_{j \in [n]\ i} c_j \cdot x_{\sigma(j)}]$ evaluates to 1 for exactly one value of x_n. Thus we continue from above and get

$$\delta(f, g) = 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} = 1 - \frac{1}{|\mathbb{F}|}.$$

□

Lemma 1.2.4 (Distance formula)**.** *Let $f, g : \mathbb{F}_q^n \to \mathbb{F}_q$. Let $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$ and $\{\gamma_c\}_{c \in \mathbb{F}_q^\times}$ be their Fourier expansions. Then*

$$\delta(f, g) = 1 - \frac{1}{q} \left(1 + \sum_{c \in \mathbb{F}_q^\times} \langle \varphi_c, \gamma_c \rangle \right).$$

Proof. By Theorem 1.1.14,

$$\Pr_x[f(x) = g(x)] = \mathbb{E}_x \left[\frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(f(x) - g(x)))} \right].$$

The term $c = 0$ contributes $1/q$. For $c \neq 0$,

$$\omega_p^{\mathrm{Tr}(c(f(x) - g(x)))} = \varphi_c(x) \overline{\gamma_c(x)}.$$

Therefore

$$\Pr_x[f(x) = g(x)] = \frac{1}{q}\left(1 + \sum_{c\in\mathbb{F}_q^\times} \langle \varphi_c, \gamma_c\rangle\right),$$

which gives the first formula after using $\delta(f,g) = 1 - \Pr_x[f(x) = g(x)]$. $\square$

Lemma 1.2.5 (Fourier expansion of a linear function)**.** *Fix $\alpha \in \mathbb{F}_q^n$ and let $\{\lambda_c\}_{c\in\mathbb{F}_q^\times}$ be the Fourier expansion of ℓ_α. Then*

$$\lambda_c = \chi_{c\alpha}.$$

Proof. For every $x \in \mathbb{F}_q^n$,

$$\lambda_c(x) = \omega_p^{\mathrm{Tr}(c\langle\alpha,x\rangle)} = \omega_p^{\mathrm{Tr}(\langle c\alpha,x\rangle)} = \chi_{c\alpha}(x).$$

$\square$

Definition 1.2.6 (Linear Fourier score)**.** For $f : \mathbb{F}_q^n \to \mathbb{F}_q$ and $\alpha \in \mathbb{F}_q^n$, define the linear Fourier score of f at α by

$$S_\alpha(f) := \sum_{c\in\mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha),$$

where $\{\varphi_c\}_{c\in\mathbb{F}_q^\times}$ is the Fourier expansion of f.

Lemma 1.2.7 (Distance to linearity in Fourier form)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$ have Fourier expansion $\{\varphi_c\}_{c\in\mathbb{F}_q^\times}$. Then*

$$\delta(f, \mathcal{LIN}) = 1 - \frac{1}{q}\left(1 + \max_{\alpha\in\mathbb{F}_q^n} S_\alpha(f)\right).$$

Moreover, for every $\alpha \in \mathbb{F}_q^n$, the score $S_\alpha(f)$ is real and satisfies $-1 \le S_\alpha(f) \le q-1$.

Proof. Apply Theorem 1.2.4 with $g = \ell_\alpha$ and use Theorem 1.2.5. This gives

$$\delta(f, \ell_\alpha) = 1 - \frac{1}{q}\left(1 + S_\alpha(f)\right).$$

Taking the minimum over α gives the claimed formula for $\delta(f, \mathcal{LIN})$. For fixed α, the same formula expresses $S_\alpha(f)$ as $q(1 - \delta(f, \ell_\alpha)) - 1$. Hence $S_\alpha(f)$ is real, and since $0 \le \delta(f, \ell_\alpha) \le 1$, we get $-1 \le S_\alpha(f) \le q-1$. $\square$

1.3 BLR linearity test

Definition 1.3.1 (Finite-field BLR test)**.** Given oracle access to $f : \mathbb{F}_q^n \to \mathbb{F}_q$, the finite-field BLR verifier $\mathsf{V}_{\mathsf{BLR}}^f$ samples $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$ independently and uniformly at random, and accepts iff

$$af(x) + bf(y) = f(ax + by).$$

That is,

$$\mathsf{V}_{\mathsf{BLR}}^f = 1 \quad\Longleftrightarrow\quad af(x) + bf(y) = f(ax+by).$$

Lemma 1.3.2 (Completeness)**.** *If $f \in \mathcal{LIN}$, then*

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] = 1.$$

Proof. If $f = \ell_\alpha$ for some $\alpha \in \mathbb{F}_q^n$, then for every $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$,

$$f(ax + by) = \langle \alpha, ax + by \rangle = a\langle \alpha, x \rangle + b\langle \alpha, y \rangle = af(x) + bf(y).$$

Thus the verifier accepts for every choice of randomness. $\square$

Lemma 1.3.3 (Exact BLR acceptance formula)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$. Then*

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] = \frac{1}{q}\left(1 + \frac{1}{(q-1)^2}\sum_{\alpha \in \mathbb{F}_q^n} S_\alpha(f)^3\right).$$

Proof. By Theorem 1.1.14, for fixed a, b, x, y,

$$\mathbb{1}[af(x) + bf(y) = f(ax + by)] = \frac{1}{q}\sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by)))}.$$

Taking expectation gives

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] = \frac{1}{q} + \frac{1}{q}\mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^\times} \varphi_{ca}(x)\varphi_{cb}(y)\varphi_{-c}(ax + by)\right].$$

Expand the three complex-valued functions into their Fourier expansions:

$$\begin{aligned}
&\mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^\times} \varphi_{ca}(x)\varphi_{cb}(y)\varphi_{-c}(ax + by)\right] \\
&= \mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^\times}\sum_{\alpha,\beta,\gamma \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\mathbb{E}_{x,y}\left[\chi_\alpha(x)\chi_\beta(y)\chi_\gamma(ax + by)\right]\right].
\end{aligned}$$

Since

$$\chi_\gamma(ax + by) = \chi_{a\gamma}(x)\chi_{b\gamma}(y),$$

Theorem 1.1.8 implies that the inner expectation is 1 exactly when

$$\alpha + a\gamma = 0 \quad \text{and} \quad \beta + b\gamma = 0,$$

and is 0 otherwise. Therefore the last display equals

$$\mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^\times}\sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(a^{-1}b\alpha)\widehat{\varphi_{-c}}(-a^{-1}\alpha)\right].$$

Substitute $\alpha = ca\eta$. Then this becomes

$$\mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^\times}\sum_{\eta \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(ca\eta)\widehat{\varphi_{cb}}(cb\eta)\widehat{\varphi_{-c}}(-c\eta)\right].$$

As a, b, c range over $\mathbb{F}_q^\times$, so do ca, cb, and $-c$. Hence this equals

$$\frac{1}{(q-1)^2}\sum_{\eta \in \mathbb{F}_q^n}\left(\sum_{d \in \mathbb{F}_q^\times} \widehat{\varphi_d}(d\eta)\right)^3 = \frac{1}{(q-1)^2}\sum_{\eta \in \mathbb{F}_q^n} S_\eta(f)^3.$$

Substituting into the expression for $\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1]$ proves the claim. $\square$

Lemma 1.3.4 (Second moment bound)**.** *For every $f : \mathbb{F}_q^n \to \mathbb{F}_q$, one has*

$$\sum_{\alpha \in \mathbb{F}_q^n} S_\alpha(f)^2 \le (q-1)^2.$$

Proof. Expanding the square gives

$$\sum_{\alpha \in \mathbb{F}_q^n} S_\alpha(f)^2 = \sum_{a,b \in \mathbb{F}_q^\times} \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha).$$

For fixed $a, b \in \mathbb{F}_q^\times$, Cauchy-Schwarz and Theorem 1.1.11 give

$$\begin{aligned}\left|\sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha)\right| &\le \left(\sum_{\alpha \in \mathbb{F}_q^n} |\widehat{\varphi_a}(a\alpha)|^2\right)^{1/2} \left(\sum_{\alpha \in \mathbb{F}_q^n} |\widehat{\varphi_b}(b\alpha)|^2\right)^{1/2} \\ &= 1.\end{aligned}$$

The equality uses that multiplication by a and by b permutes $\mathbb{F}_q^n$, and that Parseval plus $|\varphi_a(x)| = |\varphi_b(x)| = 1$ for every x make both squared L_2 norms equal to 1. Summing over the $(q-1)^2$ choices of (a, b) gives the result. ☐

Theorem 1.3.5 (Soundness of the finite-field BLR test)**.** *For every function $f : \mathbb{F}_q^n \to \mathbb{F}_q$, one has*

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] \le 1 - \delta(f, \mathcal{LIN}).$$

Proof. Let $S_\alpha(f)$ be as in Theorem 1.2.6, and let

$$M := \max_{\alpha \in \mathbb{F}_q^n} S_\alpha(f).$$

By Theorem 1.2.7,

$$1 - \delta(f, \mathcal{LIN}) = \frac{1}{q}(1 + M).$$

By Theorem 1.2.7, each $S_\alpha(f)$ is real and $S_\alpha(f) \le M$. Therefore $S_\alpha(f)^3 \le M S_\alpha(f)^2$ for every α, because $S_\alpha(f)^2 \ge 0$. Using Theorem 1.3.3 and Theorem 1.3.4,

$$\begin{aligned}\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] &= \frac{1}{q}\left(1 + \frac{1}{(q-1)^2} \sum_{\alpha \in \mathbb{F}_q^n} S_\alpha(f)^3\right) \\ &\le \frac{1}{q}\left(1 + \frac{M}{(q-1)^2} \sum_{\alpha \in \mathbb{F}_q^n} S_\alpha(f)^2\right) \\ &\le \frac{1}{q}(1 + M) \\ &= 1 - \delta(f, \mathcal{LIN}).\end{aligned}$$

☐

Chapter 2

Gowers Hatami theorem

2.1 Preliminaries

Definition 2.1.1 (σ inner product.)**.** The σ inner product between matrices A and B is defined as

$$\langle A, B\rangle_\sigma = \mathrm{Tr}(A^* B\sigma), \tag{2.1}$$

where $\sigma \geq 0$ and A^* denotes conjugate-transpose.

Definition 2.1.2 ((ϵ, σ)-representation.)**.** Given a finite group G, an integer $d \geq 1, \varepsilon \geq 0$, and a d-dimensional positive semidefinite matrix σ with trace 1 , an (ε, σ)-representation of G is a map $f : G \to U_d(\mathbb{C})$, to the $d \times d$ unitary matrices, such that

$$\mathbb{E}_{x,y\in G}\Re\left(\langle f(x)^* f(y), f\left(x^{-1}y\right)\rangle_\sigma\right) \geq 1 - \varepsilon, \tag{2.2}$$

where $\Re$ denotes taking real part of the inner product.

Proposition 2.1.3 (σ-norm Square)**.** *Let σ be positive semidefinite and let A, B be matrices of the same dimension. Then*

$$\|A - B\|_\sigma^2 = \|A\|_\sigma^2 + \|B\|_\sigma^2 - 2\,\Re\langle A, B\rangle_\sigma.$$

Proof. Expanding by linearity of the inner product,

$$\|A - B\|_\sigma^2 = \Re\langle A - B, A - B\rangle_\sigma = \|A\|_\sigma^2 + \|B\|_\sigma^2 - \Re\langle A, B\rangle_\sigma - \Re\langle B, A\rangle_\sigma.$$

Since σ is Hermitian, $\langle B, A\rangle_\sigma = \overline{\langle A, B\rangle_\sigma}$, so the two cross terms have the same real part, yielding $-2\,\Re\langle A, B\rangle_\sigma$. □

2.2 Representation theory

In general, a representation of a group G on a vector space V over a field k is a homomorphism from G to the linear endomorphisms of V. In mathlib this is the abbreviation `Representation`, defined as $G \to^* (V \to_k V)$, i.e. a monoid homomorphism into the k-linear endomorphisms of V.

We call a representation *unitary* when its image lies in the unitary group; in the finite-dimensional matrix setting used below this means a homomorphism $\rho : G \to U(V)$. In the Lean development this is represented directly as a monoid homomorphism into a matrix unitary group, for example $G \to^*$ `Matrix.unitaryGroup`.

Definition 2.2.1 (Regular representation)**.** Let $R : G \to \mathbb{C}[G]$ be a regular representation where $\{|g\rangle, g \in G\}$ forms a basis of $\mathbb{C}[G]$. The representation is given by

$$R(x)|g\rangle = |xg\rangle. \tag{2.3}$$

It is possible to decompose any representation of a finite group into direct sums over irreps.

Theorem 2.2.2 (Maschke's theorem)**.** *Let G be a finite group and let $\rho : G \to \mathrm{GL}(V)$ be a finite-dimensional representation over $\mathbb{C}$. Then ρ is completely reducible. In particular, there exists a decomposition*

$$\rho \cong \bigoplus_{m} r_m \, \rho_m, \tag{2.4}$$

where $\{\rho_m\}$ is a set of inequivalent irreducible representations of G, and $r_m \in \mathbb{Z}^+$ are the multiplicities.

Proof sketch. Maschke's theorem is partially formalized in `Mathlib.RepresentationTheory.Maschke` but incomplete. Filippo Berta, Jonathan Conrad, and Younous Malliefer are formalizing this. ☐

Proposition 2.2.3 (Decomposition of the regular representation)**.** *Let $R : G \to \mathrm{GL}(\mathbb{C}[G])$ be the regular representation of a finite group G. Then*

$$R \cong \bigoplus_{\rho \in \widehat{G}} d_\rho \, \rho, \tag{2.5}$$

where $\widehat{G}$ is a complete set of inequivalent irreducible representations of G, and $d_\rho = \dim(V_\rho)$ is the dimension of ρ.

Proof sketch. This can be proven using Maschke's theorem in 2.2.2 to decompose into irreps with multiplicity. Then use character orthogonality theorem in mathlib `FDRep.char_orthonormal` to show the multiplicity equals irrep dimension. A more formal statement is given by the Peter-Weyl theorem. ☐

Proposition 2.2.4 (Character of regular representation.)**.** *Let G be a finite group and $\sum_\rho$ denotes summing over the complete set of inequivalent irreps $\hat{G}$*

$$\sum_{\rho \in \hat{G}} d_\rho \, \mathrm{Tr}(\rho(x)) = |G| \delta_{x,e}. \tag{2.6}$$

Proof sketch. The proof uses decomposition of regular representation and definition of character. ☐

Definition 2.2.5 (Group fourier transform)**.** Let $f : G \to U_d(\mathbb{C})$ be a map and $\rho : G \to U_{d_\rho}(\mathbb{C})$ be a unitary irrep. The fourier transform of f at the representation ρ is denoted by $\hat{f}$:

$$\hat{f}(\rho) = \mathbb{E}_{x \in G} f(x) \otimes \overline{\rho(x)}, \tag{2.7}$$

where $\overline{\rho(x)}$ denotes complex conjugate of $\rho(x)$.

2.3 Gowers Hatami Theorem

Theorem 2.3.1 (Gowers-Hatami)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $f : G \to U_d(\mathbb{C})$ an (ε, σ)-representation of G. Then there exists a $d' \geq d$, an isometry $V : \mathbb{C}^d \to \mathbb{C}^{d'}$, and a representation $g : G \to U_{d'}(\mathbb{C})$ such that*

$$\mathbb{E}_{x \in G} \|f(x) - V^* g(x) V\|_\sigma^2 \leq 2\varepsilon. \tag{2.8}$$

Proof sketch. The theorem is due to Gowers and Hatami [GH17]; the constructive proof below follows Vidick's lecture notes, Theorem 2.3 [Vid22]. In particular, the proof is constructive. The isometry is defined as $V : \mathbb{C}^d \to \mathbb{C}^d \otimes \left(\oplus_\rho \mathbb{C}^{d_\rho} \otimes \mathbb{C}^{d_\rho}\right)$ by

$$Vu = \bigoplus_\rho d_\rho^{1/2} \sum_{i=1}^{d_\rho} \left(\hat{f}(\rho)\,(u \otimes e_i)\right) \otimes e_i. \tag{2.9}$$

And the exact representation is defined as

$$g(x) = \bigoplus_\rho \left(I_d \otimes I_{d_\rho} \otimes \rho(x)\right). \tag{2.10}$$

The proof requires the following two ingredients:

1. V is an isometry:
$$V^* V = \mathbb{I}_d \tag{2.11}$$

2. The isometry "averages" over group multiplication. For any $x \in G$,
$$V^* g(x) V = \mathbb{E}_z f(z)^* f(zx). \tag{2.12}$$

Both ingredients follow from the key identity over the regular representation in Eq. (2.6). □

Next, we will give a different proof that does not directly decompose into irreps but instead uses the action of the regular representation.

Lemma 2.3.2 (Gowers-Hatami Pair-Correlation)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $\rho : G \to U(\mathcal{H})$ an (ε, σ)-representation of G on space $\mathcal{H}$. Define the pair-correlation of ρ as:*

$$\mathcal{C}(\rho) := \mathbb{E}_{x,y \in G} \Re \langle \rho(y)\rho(x), \rho(yx) \rangle_\sigma. \tag{2.13}$$

Then there exists an isometry $V : \mathcal{H} \to \mathcal{H}'$ to a different space $\mathcal{H}'$, and an exact representation $\rho' : G \to U(\mathcal{H}')$ such that the average correlation between ρ and ρ' is exactly the pair-correlation, which is bounded by $1 - \varepsilon$:

$$\mathbb{E}_{x \in G} \Re \langle \rho(x), V^* \rho'(x) V \rangle_\sigma = \mathcal{C}(\rho) \geq 1 - \varepsilon. \tag{2.14}$$

Proof. Here $\mathcal{H}' = L(G, \mathcal{H})$ is the space of functions $f : G \to \mathcal{H}$.

Let the isometry be defined by the action of the approximate representation on a state $u \in \mathcal{H}$ as

$$\begin{aligned} V :& \quad \mathcal{H} \to L(G, \mathcal{H}), & (2.15)\\ V(u) :& \quad G \to \mathcal{H}, \quad x \mapsto \rho(x)(u), \quad \forall x \in G. & (2.16) \end{aligned}$$

The adjoint map V^* is defined as the unique map such that for all $u \in \mathcal{H}$ and $f \in L(G, \mathcal{H})$,

$$\langle V(u), f \rangle_{L(G,\mathcal{H})} = \langle u, V^*(f) \rangle_{\mathcal{H}}. \tag{2.17}$$

In particular, V^* is explicitly given by the group average:

$$V^* : \quad L(G, \mathcal{H}) \to \mathcal{H}, \tag{2.18}$$

$$V^*(f) \quad = \mathbb{E}_{x\in G}\left[\rho^*(x) f(x)\right]. \tag{2.19}$$

The exact representation is given by the right regular representation on $L(G, \mathcal{H})$:

$$\rho' : \quad G \to U(L(G, \mathcal{H})), \tag{2.20}$$

$$(\rho'(x)f)(y) \quad = f(yx), \quad \forall x, y \in G. \tag{2.21}$$

V is an isometry because $\rho(x)$ is unitary, yielding:

$$V^*V(u) = \mathbb{E}_{x\in G}\left[\rho^*(x)\rho(x)(u)\right] = \mathbb{E}_{x\in G}[u] = u. \tag{2.22}$$

The compression of the exact representation "averages" over group multiplication. For any $u \in \mathcal{H}$:

$$\begin{aligned} V^*\rho'(x)V(u) &= \mathbb{E}_{y\in G}\left[\rho^*(y)(\rho'(x)\rho(y))(u)\right] \\ &= \mathbb{E}_{y\in G}\left[\rho^*(y)\rho(yx)(u)\right]. \end{aligned} \tag{2.23}$$

Substituting the average property from Eq. 2.23 into the inner product, and using the unitarity of $\rho(y)$ to rewrite the adjoints ($\langle A, B^*C\rangle_\sigma = \langle BA, C\rangle_\sigma$), we obtain:

$$\begin{aligned} \mathbb{E}_{x\in G}\Re\langle\rho(x), V^*\rho'(x)V\rangle_\sigma &= \mathbb{E}_{x,y\in G}\Re\langle\rho(x), \rho^*(y)\rho(yx)\rangle_\sigma \\ &= \mathbb{E}_{x,y\in G}\Re\langle\rho(y)\rho(x), \rho(yx)\rangle_\sigma \\ &= \mathcal{C}(\rho). \end{aligned} \tag{2.24}$$

By the definition of an (ε, σ)-approximate representation, the pair-correlation inherently satisfies $\mathcal{C}(\rho) \geq 1 - \varepsilon$, which concludes the proof. □

Theorem 2.3.3 (Gowers-Hatami-2)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $\rho : G \to U(\mathcal{H})$ an (ε, σ)-representation of G on space $\mathcal{H} = \mathbb{C}^d$. Then there exists an isometry $V : \mathcal{H} \to \mathcal{H}'$ to a different space $\mathcal{H}'$, and a representation $\rho' : G \to \mathcal{H}'$ such that*

$$\mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \leq 2\varepsilon. \tag{2.25}$$

Proof. A regular-representation proof of the Gowers-Hatami theorem appears in Morel's slides [Mor20], following the original Gowers-Hatami theorem [GH17].

The rest of the proof follows by expanding the σ-norm, bounding the norm of $\mathbb{E}_{x\in G}V^*\rho'(x)V$ and applying the definition of (ε, σ)-representation. In particular, we have

$$\begin{aligned} &\mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \\ = \ &\mathbb{E}_{x\in G}\|\rho(x)\|_\sigma^2 + \mathbb{E}_{x\in G}\|V^*\rho'(x)V\|_\sigma^2 - 2\mathbb{E}_{x\in G}\Re\langle\rho(x), V^*\rho'(x)V\rangle_\sigma \end{aligned} \tag{2.26}$$

Because ρ is unitary and $\mathrm{Tr}\sigma = 1$, $\mathbb{E}_{x\in G}\|\rho(x)\|_\sigma^2 = 1$. For the second term, recall from Eq. (2.23) that the compression is the group average $V^*\rho'(x)V = \mathbb{E}_{y\in G}\rho^*(y)\rho(yx)$. Each summand $\rho^*(y)\rho(yx)$ is a product of unitaries, hence unitary, so $\|\rho^*(y)\rho(yx)\|_\sigma = 1$. By convexity of the σ-norm—the triangle inequality applied to the average—we obtain

$$\|V^*\rho'(x)V\|_\sigma = \left\|\mathbb{E}_{y\in G}\rho^*(y)\rho(yx)\right\|_\sigma \leq \mathbb{E}_{y\in G}\|\rho^*(y)\rho(yx)\|_\sigma = 1, \tag{2.27}$$

hence $\mathbb{E}_{x\in G}\|V^*\rho'(x)V\|_\sigma^2 \leq 1$.

Therefore using the bounds in Eq. 2.27 and Eq. 2.24 by applying Lemma 2.3.2, we have

$$\mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \leq 2 - 2(1 - \varepsilon) = 2\varepsilon. \tag{2.28}$$

□

2.4 Gowers-Hatami for BLR linearity test over $\mathbb{F}_q$

Assume $q = p^s$ be a prime with power $s \in \mathbb{Z}^+$, $n \geq 1$, and we set

$$X := \mathbb{F}_q^n, \qquad A := \mathbb{F}_q^\times, \qquad \omega := e^{2\pi i/p}.$$

First notice that linearity implies two properties:

1. Additivity $f(x+y) = f(x) + f(y), \quad \forall x, y \in X$.

   Note that for q prime additivity is actually equivalent to linearity. However this is not the case for prime powers $q = p^s$ for $s > 1$.

2. Homogeneity $f(ax) = af(x), \quad \forall x \in X,\ a \in A$.

   For $q = 2$, the two properties are redundant. For $q > 2$, turns out testing homogeneity is crucial for a test to have a high soundness.

2.4.1 The twisted group G

Define a group G by the set $X \times A$ with the binary operation

$$(x, a) \star (y, b) := (b^{-1}x + a^{-1}y,\ ab). \tag{2.29}$$

Lemma 2.4.1 (G is an abelian group)**.** *The operation $\star$ in (2.29) makes G into an abelian group with identity $e = (0, 1)$ and inverses $(x, a)^{-1} = (-a^2x, a^{-1})$.*

Proof. Associativity. For $(x, a), (y, b), (z, c) \in G$,

$$((x, a) \star (y, b)) \star (z, c) = (b^{-1}x + a^{-1}y,\ ab) \star (z, c) = (b^{-1}c^{-1}x + a^{-1}c^{-1}y + a^{-1}b^{-1}z,\ abc),$$

and an identical computation gives the same expression for $(x, a) \star ((y, b) \star (z, c))$.

Commutativity. $(y, b) \star (x, a) = (a^{-1}y + b^{-1}x,\ ba) = (b^{-1}x + a^{-1}y,\ ab) = (x, a) \star (y, b)$.

Identity. For any $(y, b) \in G$, $(0, 1) \star (y, b) = (b^{-1} \cdot 0 + 1^{-1} \cdot y,\ 1 \cdot b) = (y, b)$.

Inverse. For any $(x, a) \in G$,

$$(x, a) \star (-a^2x, a^{-1}) = ((a^{-1})^{-1} \cdot x + a^{-1} \cdot (-a^2x),\ a \cdot a^{-1}) = (ax - ax,\ 1) = (0, 1).$$

$\square$

It is at first surprising to come up with $\star$ operation, but G is isomorphic to the direct product group.

Remark 2.4.2 (Group isomorphism to product group). The map

$$\Psi : G \to (X, +) \times (A, \cdot), \qquad \Psi(x, a) := (ax, a), \tag{2.30}$$

is a group isomorphism onto the direct product of $(X, +)$ and $(A, \cdot)$ with operation $\star_\Psi : (x,\ a) \star_\Psi (y,\ b) = (x + y,\ ab)$. Indeed,

$$\Psi((x, a) \star (y, b)) = (ab \cdot (b^{-1}x + a^{-1}y),\ ab) = (ax + by,\ ab) = \Psi(x, a) \star_\Psi \Psi(y, b),$$

and Ψ is bijective with inverse $\Psi^{-1}(y, b) = (b^{-1}y, b)$.

This isomorphism is used only to *motivate* the characters below. The Lean development does not formalize Ψ: it equips G with its group structure (Lemma 2.4.1) and defines the characters $\chi_{\alpha,k}$ (Proposition 2.4.4) directly on the twisted group, proving the homomorphism property by direct computation rather than by pulling back through Ψ. Accordingly it is recorded as a remark, with no `\lean` link and no node in the dependency graph.

The isomorphism gives us a easy way to compute its irreducible representations which forms the Pontryagin dual group (the set of irrep characters for an abelian group).

Remark 2.4.3 (Pontryagin dual over direct product). For finite abelian groups G_1, G_2 with direct product $H = G_1 \times G_2$, the Pontryagin dual (the set of irreducible characters) factors as $\hat{H} = \hat{G}_1 \times \hat{G}_2$. Like the isomorphism above, this is used only to motivate the character formula and is not part of the formalized graph.

Proposition 2.4.4 (Group characters)**.** *The group $(G, \star)$ has the Pontryagin dual character group*

$$\hat{G} = \{\chi_{\alpha,k}\}_{\alpha \in X, k \in [0,\cdots,q-2]}, \quad \chi_{\alpha,k}(x,a) = \omega^{\mathrm{Tr}\langle \alpha, ax\rangle}\psi_k(a), \tag{2.31}$$

where $\psi_k(a) = e^{\frac{i2\pi mk}{q-1}}$ and $a = g_0^m$ for a fixed multiplicative generator $g_0 \in A$.

Proof. By the isomorphism $\Psi(x,a) = (ax, a)$ in the remark above, $(G,\star) \cong (X,+) \times (A,\cdot)$. First we consider the characters of the product group. Using the Pontryagin-dual factorization in the remark above, we have

$$\widehat{(X,+) \times (A,\cdot)} = \hat{X} \times \hat{A}.$$

The character groups of X and A are

$$\hat{X} = \{\phi_\alpha(y) = \omega^{\mathrm{Tr}\langle \alpha, y\rangle}\}_{\alpha \in X}, \tag{2.32}$$

$$\hat{A} = \{\psi_k(b) = e^{\frac{i2\pi mk}{q-1}}\}_{k \in [0,\cdots,q-2]}, \tag{2.33}$$

where $b = g_0^m$ for a fixed multiplicative generator $g_0 \in A$. Here the field trace $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is defined by $\mathrm{Tr}(x) = x + x^p + x^{p^2} + \cdots + x^{p^{s-1}}$ for $x \in \mathbb{F}_q$ (see Definition 1.1 in Chapter 1).Therefore, $\widehat{(X,+) \times (A,\cdot)} = \{\Theta_{\alpha,k}(y,b) = \phi_\alpha(y)\psi_k(b)\}_{\alpha,k}$. The character $\chi \in \hat{G}$ is given by the pull-back through the isomorphism

$$\chi_{\alpha,k}(x,a) = \Theta_{\alpha,k}(\Psi(x,a)) = \Theta_{\alpha,k}(ax, a). \tag{2.34}$$

Therefore, we have $\chi_{\alpha,k}(x,a) = \phi_\alpha(ax)\psi_k(a) = \omega^{\mathrm{Tr}\langle \alpha, ax\rangle}\psi_k(a)$. □

2.4.2 Approximate representation and matrix lift of f

Definition 2.4.5 (Matrix lift F_f)**.** Define a unitary matrix valued lift $F_f : G \to U(\mathbb{C}^{q-1})$ by the diagonal matrix

$$F_f(x,a) = D(af(x)) := \sum_{c \in A} \omega^{\mathrm{Tr}\, caf(x)} |c\rangle\langle c|, \tag{2.35}$$

where $\{|c\rangle\}$ denotes the orthonormal basis vectors of a Hilbert space $\mathcal{H} \cong \mathbb{C}^{q-1}$ and the inner product between operators defined by $\langle A, B\rangle_\sigma = \mathrm{Tr}(A^* B \sigma)$ and choose $\sigma = \mathbb{I}/(q-1)$.

We first show that the BLR condition indeed gives an approximate represenatation.

Lemma 2.4.6 (Lift operation from BLR condition)**.** *Let $f : X \to \mathbb{F}_q$ and $(x,a),(y,b) \in G$ be rejected by the BLR test with probability ε, then the unitary matrix lift F_f in Equation 2.35 is an $(\frac{q}{q-1}\varepsilon, \frac{\mathbb{I}}{q-1})$ approximate representation of $(G,\star)$.*

Proof. The BLR test condition gives

$$\Pr_{c,d,x,y}\big[cf(x) + df(y) \neq f(cx + dy)\big] \leq \varepsilon.$$

First we observe that

$$\Pr_{g,h\in G}[F_f(g\star h) = F_f(g)F_f(h)] = \Pr_{c,d,x,y}[cf(x)+df(y) = f(cx+dy)]. \tag{2.36}$$

Using Equation 2.29 and Equation 2.35,

$$F_f((x,a)\star(y,b)) = \sum_{c\in A}\omega^{\operatorname{Tr} cab\, f(b^{-1}x+a^{-1}y)}|c\rangle\langle c|,$$
$$F_f(x,a)\,F_f(y,b) = \sum_{c\in A}\omega^{\operatorname{Tr} c(af(x)+bf(y))}|c\rangle\langle c|.$$

These coincide iff their exponents agree modulo p for $c\in A$:

$$ab\, f(b^{-1}x+a^{-1}y) = af(x)+bf(y) \mod p, \quad \forall c\in A.$$

Therefore, we have equality over $\mathbb{F}_p$

$$ab\, f(b^{-1}x+a^{-1}y) = af(x)+bf(y).$$

Since a,b have a multiplicative inverse, substitute $c := b^{-1}$, $d := a^{-1}$, we have

$$f(cx+dy) = cf(x)+df(y),$$

which proves Equation 2.36.

To relate the test condition to the approximate representation, first note the equivalent definition of approximate representation

$$\mathbb{E}_{g,h\in G}\Re\left\langle F_f(g)^*F_f(h), F_f\left(g^{-1}h\right)\right\rangle_\sigma = \mathbb{E}_{g,h\in G}\Re\left\langle F_f(g)F_f(h), F_f\left(gh\right)\right\rangle_\sigma, \tag{2.37}$$

because F_f is unitary so $F_f(g)^* = F_f(g^{-1}), \forall g\in G$. Then evaluate the right hand side

$$\begin{aligned}\mathbb{E}_{g,h\in G}\Re\left\langle F_f(g)F_f(h), F_f\left(gh\right)\right\rangle_\sigma &= \mathbb{E}_{g,h\in G}\Re\operatorname{Tr}\left[F_f^*(h)F_f^*(g)F_f\left(gh\right)\sigma\right],\\ &= \mathbb{E}_{(x,a),(y,b)\in G}\frac{1}{q-1}\Re\operatorname{Tr}\left[\sum_{c\in A}\omega^{-\operatorname{Tr} c(af(x)+bf(y))}\omega^{\operatorname{Tr} cab\, f(b^{-1}x+a^{-1}y)}|c\rangle\langle c|\right],\\ &= \mathbb{E}_{(x,a),(y,b)\in G}\frac{1}{q-1}\Re\sum_{c\in A}\omega^{\operatorname{Tr} cz}, \end{aligned} \tag{2.38}$$

where we define $z = c(ab\, f(b^{-1}x+a^{-1}y) - (af(x)+bf(y)))$. To compute the expectation, we can split to two cases

$$\sum_{c\in A}\omega^{\operatorname{Tr} cz} = \begin{cases} q-1, & z=0 \quad \text{when } f \text{ passes BLR},\\ -1, & z\neq 0 \quad \text{when } f \text{ fails BLR}.\end{cases} \tag{2.39}$$

The second case is because $\sum_{c\in A}\omega^{\operatorname{Tr} cz}+1 = \sum_{c\in\mathbb{F}_q}\omega^{\operatorname{Tr} cz} = 0$. Therefore, we have the approximate representation

$$\mathbb{E}_{g,h\in G}\Re\left\langle F_f(g)F_f(h), F_f\left(gh\right)\right\rangle_\sigma \geq (1-\varepsilon) - \frac{1}{q-1}\varepsilon = 1 - \frac{q}{q-1}\varepsilon. \tag{2.40}$$

Hence F_f is an approximate representation with $\epsilon = \frac{q}{q-1}\varepsilon$ as claimed.

□

The approximate representation "appears" slightly worse than what we would have hoped $\epsilon > \varepsilon$, but we will see this is not an issue for achieving a high soundness. Remarkably, the twisted group operation let us recover the linearity test condition exactly.

2.4.3 Gowers-Hatami implies BLR high soundness

In this section, we first prove the key lemma relating the implication of GH theorem to the hamming distance. The proof outlined here correspond to the formalized version in lean. The most convenient proof still uses properties of the Fourier transform of F_f. However, we remark that there are other alternative proofs, which we leave as comment in the current source file.

First let us write down statements about Fourier coefficients of F_f, which we will use in the proof.

Proposition 2.4.7 (Fourier coefficient of F_f)**.** *Let the matrix lift $F_f : G \to U(\mathbb{C}^{q-1})$ be defined by the diagonal matrix*

$$F_f(x,a) = D(af(x)) := \sum_{c\in A} \omega^{\operatorname{Tr} caf(x)} |c\rangle\langle c|,$$

then its Fourier coefficients can be written as

$$[\hat{F}_f(\chi_{c\beta,k})]_{cc} = \psi_k(c)\nu_k(\beta), \tag{2.41}$$

where

$$\nu_k(\beta) := \mathbb{E}_{x,b}\left[\omega^{\operatorname{Tr} b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\right], \tag{2.42}$$

Proof. By Proposition 2.4.4, the characters are given by

$$\hat{G} = \{\chi_{\alpha,k}\}_{\alpha\in X, k\in[0,\cdots,q-2]}, \chi_{\alpha,k}(x,a) = \omega^{\operatorname{Tr}\langle\alpha,ax\rangle}\psi_k(a).$$

Therefore the Fourier coefficients of F_f are

$$\begin{aligned}\hat{F}_f(\chi_{\alpha,k}) &= \sum_{c\in\mathbb{F}_q^\times} \mathbb{E}_{x,a}\omega^{\operatorname{Tr} caf(x)}\bar{\chi}_{\alpha,k}(x,a), \\ &= \sum_{c\in\mathbb{F}_q^\times} \mathbb{E}_{x,a}\left[\omega^{\operatorname{Tr} a(cf(x)-\langle\alpha,x\rangle)}\overline{\psi_k(a)}\right] |c\rangle\langle c|. \end{aligned} \tag{2.43}$$

Since $c \in \mathbb{F}_q^\times$ is invertible, we perform a change of variables. Let $\alpha = c\beta$ for a fresh $\beta \in \mathbb{F}_q^n$. Because c is fixed, this is a bijection on $\mathbb{F}_q^n$. We also substitute $b = ca$, which is a bijection on $\mathbb{F}_q^\times$. This means $a = bc^{-1}$, and by the multiplicity of Dirichlet characters, $\overline{\psi_k(a)} = \overline{\psi_k(bc^{-1})} = \overline{\psi_k(b)}\psi_k(c)$.

Substituting these into the expectation isolates the c-dependence:

$$\begin{aligned}[\hat{F}_f(\chi_{c\beta,k})]_{cc} &= \mathbb{E}_{x,b}\left[\omega^{\operatorname{Tr} b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\psi_k(c)\right] \\ &= \psi_k(c)\nu_k(\beta), \end{aligned} \tag{2.44}$$

where $\nu_k(\beta) := \mathbb{E}_{x,b}\left[\omega^{\operatorname{Tr} b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\right]$ is a c-independent coefficient. □

Lemma 2.4.8 (Gowers-Hatami to linearity (formalized proof))**.** *Suppose $F_f : G \to U(\mathbb{C}^{q-1})$ defined in Equation 2.35 via $f : \mathbb{F}_q^n \to \mathbb{F}_q$ is an (ϵ,σ) approximate unitary representation of $(G,\star)$ defined in Equation 2.29. Implicitly, $\epsilon \le 1$.*

Then, there exists a linear function $\ell : \mathbb{F}_q^n \to \mathbb{F}_q$ such that the relative Hamming distance $\delta(f,\ell)$ is bounded by:

$$\delta(f,\ell) \le \frac{q-1}{q}\epsilon. \tag{2.45}$$

Proof. Write the diagonal lift as $F_f(g) = \mathrm{diag}_{c\in\mathbb{F}_q^\times}(\varphi_c(g))$, where the scalar functions are $\varphi_c(x,a) = \omega^{\mathrm{Tr}(caf(x))}$.

Because F_f is diagonal and $\sigma = \mathbb{I}/(q-1)$, the trace equates to the average over the diagonal elements. By an intermediate statement in the proof of Gowers-Hatami theorem using Lemma. 2.3.2, the compression from an exact representation can be rewritten as a pair-correlation:

$$\mathcal{C}(f) := \mathbb{E}_{g,h}\Re\langle F_f(g)F_f(h), F_f(gh)\rangle_\sigma \geq 1-\epsilon. \tag{2.46}$$

To connect the operator pair-correlation directly to the cubic trace of the Fourier coefficients, we express the lift matrices using the inverse Fourier transform, $F_f(x) = \sum_{\chi\in\widehat{G}}\hat{F}_f(\chi)\chi(x)$. Substituting this into the inner product definition of the pair-correlation and using the character homomorphism $\chi(gh) = \chi(g)\chi(h)$, we expand the trace:

$$\begin{aligned}\mathcal{C}(f) &= \mathbb{E}_{g,h\in G}\Re\mathrm{Tr}\left(F_f(g)^*F_f(h)^*F_f(gh)\sigma\right)\\ &= \Re\mathrm{Tr}\left(\sum_{\chi_1,\chi_2,\chi_3}\hat{F}_f^*(\chi_1)\hat{F}_f^*(\chi_2)\hat{F}_f(\chi_3)\sigma\cdot\mathbb{E}_g\left[\overline{\chi_1(g)}\chi_3(g)\right]\mathbb{E}_h\left[\overline{\chi_2(h)}\chi_3(h)\right]\right).\end{aligned} \tag{2.47}$$

By the orthogonality of characters, the expectations over g and h independently vanish unless $\chi_1 = \chi_3$ and $\chi_2 = \chi_3$. This completely collapses the triple sum to a single sum over χ. Because the Fourier coefficients of the diagonal lift F_f are themselves diagonal matrices (and thus commute), we can rearrange the surviving terms to recover the exact cubic trace expression:

$$\mathcal{C}(f) = \Re\sum_{\chi\in\widehat{G}}\mathrm{Tr}\left(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma\right). \tag{2.48}$$

Because the Fourier coefficients are all fully diagonal since F_f is diagonal, we expand the product

$$\sum_{\chi_{\alpha,k}\in\widehat{G}}\mathrm{Tr}(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma) = \frac{1}{q-1}\sum_{\alpha\in\mathbb{F}_q^n}\sum_{k=0}^{q-2}\sum_{c\in\mathbb{F}_q^\times}\left|[\hat{F}_f(\chi_{\alpha,k})]_{cc}\right|^2\overline{[\hat{F}_f(\chi_{\alpha,k})]_{cc}}.$$

Using the substitution $\alpha = c\beta$, summing over all α is equivalent to summing over all β. Substituting our factored coefficient from Eq. (2.41) in Proposition 2.4.7:

$$\begin{aligned}\sum_{\chi\in\widehat{G}}\mathrm{Tr}(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma) &= \frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\sum_{\beta\in\mathbb{F}_q^n}\sum_{k=0}^{q-2}|\psi_k(c)\nu_k(\beta)|^2\overline{\psi_k(c)\nu_k(\beta)}\\ &= \frac{1}{q-1}\sum_{\beta,k}|\nu_k(\beta)|^2\overline{\nu_k(\beta)}\sum_{c\in\mathbb{F}_q^\times}\overline{\psi_k(c)},\end{aligned}$$

where

$$\nu_k(\beta) := \mathbb{E}_{x,b}\left[\omega^{b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\right],$$

is defined in Eq. (2.42).

Use the indicator function of $\mathbb{F}_q^\times$:

$$\sum_{c\in\mathbb{F}_q^\times}\overline{\psi_k(c)} = (q-1)\mathbb{I}(k=0)$$

we have

$$\sum_{\chi\in\widehat{G}} \mathrm{Tr}(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma) = \sum_{\beta\in\mathbb{F}_q^n} |\nu_0(\beta)|^2\overline{\nu_0(\beta)}. \tag{2.49}$$

Note that $\nu_0(\beta) \in \mathbb{R}$ is related to the hamming distance $\delta(f, \ell_\beta)$ for liner function $\ell_\beta = \langle f, \ell_\beta\rangle$ via

$$\nu_0(\beta) = (1 - \delta(f, \ell_\beta)) - \frac{1}{q-1}\delta(f, \ell_\beta) = 1 - \frac{q}{q-1}\delta(f, \ell_\beta). \tag{2.50}$$

Moreover, because the Fourier coefficients sum to identity by Eq. (**??**), we have

$$\begin{aligned}
1 &= \frac{1}{q-1}\sum_{\chi\in\hat{G}} \mathrm{Tr}\left[\hat{F}_f^*(\chi)\hat{F}_f(\chi)\right], \\
&= \frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\sum_{\beta\in\mathbb{F}_q^n}\sum_k |\psi_k(c)\nu_k(\beta)|^2, \\
&= \sum_{\beta\in\mathbb{F}_q^n}\sum_k |\nu_k(\beta)|^2.
\end{aligned} \tag{2.51}$$

Therefore $|\nu_k(\beta)|^2$ is a probability distribution indexed by (k, β). Because the $k = 0$ probability is a subset, we have

$$\sum_{\beta\in\mathbb{F}_q^n} |\nu_0(\beta)|^2 \le \sum_{\beta\in\mathbb{F}_q^n}\sum_k |\nu_k(\beta)|^2 = 1. \tag{2.52}$$

Let β^* be the closest linear function such that

$$\nu_0(\beta^*) := \max_{\beta\in\mathbb{F}_q^n} \nu_0(\beta). \tag{2.53}$$

Therefore from Gowers-Hatami implication in Eq. (2.46) and the simplification of Fourier coefficeints in Eq. (2.49), we have the bound

$$1 - \epsilon \le \Re\sum_{\chi\in\widehat{G}} \mathrm{Tr}(\hat{F}_f^*(\chi)\hat{F}_f(\chi)\hat{F}_f^*(\chi)\sigma) = \sum_{\beta\in\mathbb{F}_q^n} |\nu_0(\beta)|^2\overline{\nu_0(\beta)} \le \nu_0(\beta^*)\sum_{\beta\in\mathbb{F}_q^n} |\nu_0(\beta)|^2 \le \nu_0(\beta^*). \tag{2.54}$$

Unpack the relation to the hamming distance, we have

$$1 - \frac{q}{q-1}\delta(f, \ell_{\beta^*}) \ge 1 - \epsilon \implies \delta(f, \ell_{\beta^*}) \le \frac{q-1}{q}\epsilon. \tag{2.55}$$

□

Lemma 2.4.9 (Soundness of the BLR test)**.** *Let q be prime. For every $f : \mathbb{F}_q^n \to \mathbb{F}_q$, $\delta(f, \mathcal{LIN}) \le \Pr_{a,b,x,y}[af(x) + bf(y) \neq f(ax + by)]$.*

Alternative proof via Gowers–Hatami. First denote

$$X := \mathbb{F}_q^n, \qquad A := \mathbb{F}_q^\times, \qquad \omega := e^{2\pi i/p}.$$

Define a group G on the set $X \times A$ with the binary operation

$$(x, a) \star (y, b) := (b^{-1}x + a^{-1}y,\ ab).$$

Let F_f be the unitary matrix valued lift $F_f : G \to U(\mathbb{C}^{q-1})$ given by the diagonal matrix

$$F_f(x,a) = D(af(x)) := \sum_{c \in A} \omega^{\operatorname{Tr} caf(x)} |c\rangle\langle c|,$$

where $\{|c\rangle\}$ denotes the orthonormal basis vectors of a Hilbert space $\mathcal{H} \cong \mathbb{C}^{q-1}$, the inner product between operators is $\langle A, B\rangle_\sigma = \operatorname{Tr}(A^* B \sigma)$, and $\sigma = \mathbb{I}/(q-1)$. By Lemma 2.4.6, since f is rejected with probability ε, the lift F_f is a $(\frac{q}{q-1}\varepsilon, \frac{\mathbb{I}}{q-1})$ approximate representation of $(G, \star)$.

Then by Lemma 2.4.8, there exists a linear function $\ell : \mathbb{F}_q^n \to \mathbb{F}_q$ whose relative Hamming distance to f is bounded by

$$\delta(f,\ell) \le \frac{q-1}{q} \cdot \frac{q}{q-1}\varepsilon = \varepsilon.$$

This proves the claim

$$\delta(f, \mathcal{LIN}) := \min_\alpha \delta(f, \ell_\alpha) \le \delta(f,\ell) \le \varepsilon.$$

Miraculously, the loose factor $\frac{q}{q-1}$ in the approximate-representation parameter cancels the tighter factor $\frac{q-1}{q}$ of Lemma 2.4.8.

In Lean this route is formalized as `gh_blr_soundness_acceptance`, which proves `acceptanceProbabilityBLR f ≤ 1 - distanceToLinear f` (the statement of `blr_soundness`), valid for $\frac{q}{q-1}\varepsilon \le 1$; the unconditional statement is the direct proof above, since for $\frac{q}{q-1}\varepsilon > 1$ the third-moment extraction's $\max \ge 0$ step fails. $\square$

Chapter 3

Exponential-length PCP

Definition 3.0.1. Let $m, n > 0$ be integers, $\mathbb{F}$ be a field, $R = \mathbb{F}[x_1, \dots, x_n]$ be its polynomial ring and R_d denote the subspace of polynomials of degree $\leq d$.

$$\mathrm{QESAT}(\mathbb{F}, n) := \{(p_1, \dots, p_m) \in R_2^m \mid \exists a \in \mathbb{F}^n,\ \forall i \in [m],\ p_i(a_1, \dots, a_n) = 0\}.$$

For example, $(x+1, xy+z) \in \mathrm{QESAT}(\mathbb{F}_2, 3)$. The goal of this chapter is to construct an exponential-length PCP for $\mathrm{QESAT}(\mathbb{F}_2, n)$ (Theorem 3.7.7). Since $\mathrm{QESAT}(\mathbb{F}_2, n)$ is NP-complete, this will imply the existence of exponential-length PCPs for all languages in NP.

3.1 Oracle computations and verifiers

Instead of oracle Turing machines, we will define verifiers using Lean monads, specifically the `OracleComp` free monad from VCV-io.

Definition 3.1.1 (Oracle specification)**.** An *oracle specification* defines the type of the queries and outputs that an oracle can handle.

Definition 3.1.2. For this section, the *randomness oracle specification* will be of a monadic function that takes nothing and returns an element from a field $\mathbb{F}$.

Definition 3.1.3 (Query implementation)**.** A *query implementation* is a monadic function whose behavior conforms to a given oracle specification.

Definition 3.1.4. The *uniform sampling query implementation* for a field $\mathbb{F}$ is a (mathematical) query implementation for the randomness oracle specification that samples uniformly from $\mathbb{F}$.

This is the the oracle implementation that will be used to prove theorems about our verifiers, since it is compatible with the more classical notion of probability distributions used in the previous chapters.

Remark 3.1.5 (Implementing verifiers)*.* If we want to actually run the verifiers, we can also use concrete implementations based on the IO monad and system randomness for pseudorandom sampling.

Definition 3.1.6 (Oracle context)**.** An *oracle context* is a bundle consisting of an oracle specification and a query implementation for that specification.

Definition 3.1.7 (Randomness oracle)**.** As *randomness oracle*, we will use the uniform sampling query implementation for the randomness oracle specification.

Definition 3.1.8 (Oracle computation)**.** An *oracle computation* is a monadic computation in the free monad generated by an oracle specification, i.e. a description (a "script") of a program consisting of a chain of oracle queries (compatible with any query implementation of the specification) and deterministic Lean computations.

A *verifier* is a function that takes an input $x \in \{0,1\}^*$ and produces an oracle computation (with query access to a randomness oracle) that outputs a boolean. This definition can be extended to include other oracles (typically a proof oracle). In Lean this convention is represented directly by function types returning `OracleComp` computations, rather than by a standalone verifier declaration.

We also need to define verifiers that do not take any inputs (for example the BLR verifier). Since they use the same randomness oracle specification as the normal verifiers, they can compose nicely with them. A *unit verifier* (or just *verifier* when it is clear from the context) is an oracle computation with query access to a randomness oracle that outputs a boolean. This definition can be extended to include other oracles (typically a proof oracle).

Remark 3.1.9 (Time complexity)*.* The price to pay for using these definitions for verifiers instead of Turing machines is that we have no simple way to enforce the time complexity of the verifier. For that, a possible approach would be to restrict the allowed operations to a strict set of operations instead of general Lean computations, and track the time cost using query complexity.

3.2 PCP and LPCP definitions

Definition 3.2.1. A *PCP verifier* is a verifier with query access to a function $\pi : [\ell(|x|)] \to \Sigma$.

Definition 3.2.2. A language $L \subseteq \{0,1\}^*$ is in $\mathrm{PCP}[\varepsilon_c, \varepsilon_s, \Sigma, \ell, q, r]$ if there exists a polynomial-time PCP verifier V such that for every $x \in \{0,1\}^*$, V makes at most $q(|x|)$ queries to the proof oracle and at most $r(|x|)$ randomness queries, and the following holds:

- Completeness: If $x \in L$, then $\exists \pi \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\pi}(x) = 1\right] \geq 1 - \varepsilon_c$.
- Soundness: If $x \notin L$, then $\forall \tilde{\pi} \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\tilde{\pi}}(x) = 1\right] \leq \varepsilon_s$.

Definition 3.2.3. A *LPCP verifier* is a verifier with query access to a linear function $\langle \pi, \cdot \rangle : \Sigma^{\ell(|x|)} \to \Sigma$.

Definition 3.2.4. A language $L \subseteq \{0,1\}^*$ is in $\mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma, \ell, q, r]$ if there exists a polynomial-time LPCP verifier V such that for every $x \in \{0,1\}^*$, V makes at most $q(|x|)$ queries to the linear proof oracle and at most $r(|x|)$ randomness queries, and the following holds:

- Completeness: If $x \in L$, then $\exists \pi \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\langle \pi, \cdot \rangle}(x) = 1\right] \geq 1 - \varepsilon_c$.
- Soundness: If $x \notin L$, then $\forall \tilde{\pi} \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\langle \tilde{\pi}, \cdot \rangle}(x) = 1\right] \leq \varepsilon_s$.

3.3 Simpler BLR verifier for $\mathbb{F}_2$

Definition 3.3.1. Let $V^f_{\mathrm{BLR}(\mathbb{F}_2, n)}$ be the verifier with oracle access to $f : \mathbb{F}_2^n \to \mathbb{F}_2$ defined as follows:

1. Sample $x, y \leftarrow \mathbb{F}_2^n$.
2. Accept if and only if $f(x) + f(y) = f(x + y)$.

Theorem 3.3.2. *Let $n > 0$. For every linear function $f : \mathbb{F}_2^n \to \mathbb{F}_2$, we have*

$$\Pr\left[V^f_{\mathrm{BLR}(\mathbb{F}_2,n)} = 1\right] = 1.$$

Proof. □

Theorem 3.3.3. *Let $n > 0$ and $f : \mathbb{F}_2^n \to \mathbb{F}_2$.*

$$\Pr\left[V^f_{\mathrm{BLR}(\mathbb{F}_2,n)} = 1\right] \leq 1 - \delta(f, \mathcal{LIN}).$$

Proof. □

3.4 Linear PCP for linear equations

Definition 3.4.1. For a field $\mathbb{F}$ and parameters $m, n \in \mathbb{N}$, we define the language

$$\mathrm{LINEQ}(\mathbb{F}, m, n) := \{(M, c) \in \mathbb{F}^{m \times n} \times \mathbb{F}^m \mid \exists b \in \mathbb{F}^n,\ Mb = c\}.$$

Definition 3.4.2. Let $V^{\langle b,\cdot\rangle}_{\mathrm{LINEQ}(\mathbb{F},m,n)}(M, c)$ be the LPCP verifier defined as follows:

1. Sample $r \leftarrow \mathbb{F}^m$.
2. Query $b \in \mathbb{F}^n$ at $u := M^\top r \in \mathbb{F}^n$.
3. Accept if and only if $\langle b, u\rangle = \langle c, r\rangle$.

Lemma 3.4.3. *Let $a \in \mathbb{F}^m$ be nonzero. Then*

$$|\mathbb{F}| \cdot \Pr_{r \leftarrow \mathbb{F}^m}[\langle a, r\rangle = 0] \leq 1.$$

Equivalently, a nonzero linear form over $\mathbb{F}^m$ vanishes on a uniformly random input with probability at most $1/|\mathbb{F}|$.

Proof. Choose a coordinate i with $a_i \neq 0$. After fixing all other coordinates of r, there is exactly one value of r_i making $\langle a, r\rangle = 0$. □

Theorem 3.4.4.

$$\mathrm{LINEQ}(\mathbb{F}, m, n) \in \mathrm{LPCP}\left[\varepsilon_c = 0, \varepsilon_s = 1/|\mathbb{F}|, \Sigma = \mathbb{F}, \ell = |x|, q = 1, r = m\right].$$

Proof. Consider the LPCP verifier $V^{\langle b,\cdot\rangle}_{\mathrm{LINEQ}(\mathbb{F},m,n)}(M, c)$.
Completeness: If $Mb = c$, then for every $r \in \mathbb{F}^m$, we have

$$\langle b, u\rangle = b^\top(M^\top r) = (Mb)^\top r = \langle c, r\rangle.$$

Soundness: If $Mb \neq c$, then

$$\begin{aligned}\Pr_{r \leftarrow \mathbb{F}^m}[\langle b, u\rangle = \langle c, r\rangle] &= \Pr_{r \leftarrow \mathbb{F}^m}[\langle Mb, r\rangle = \langle c, r\rangle] \\ &= \Pr_{r \leftarrow \mathbb{F}^m}\left[\sum_{i=1}^m (Mb - c)_i \cdot r_i = 0\right] \leq 1/|\mathbb{F}|,\end{aligned}$$

where the last inequality follows from 3.4.3. □

3.5 Linear PCP for tensor checks

For a field $\mathbb{F}$, vectors $a \in \mathbb{F}^n$ and $b \in \mathbb{F}^m$, the tensor product $a \otimes b \in \mathbb{F}^{n \times m}$ is the matrix defined by

$$(a \otimes b)_{i,j} := a_i b_j \quad \forall i \in [n],\ j \in [m].$$

We write $\mathrm{flat}(a \otimes b) \in \mathbb{F}^{nm}$ for the row-concatenation of $a \otimes b$.

For our purposes, we want a test that checks whether $\mathrm{flat}(a \otimes a) = b$

Definition 3.5.1. For a field $\mathbb{F}$ and $n \in \mathbb{N}$, define the tensor test language

$$\mathrm{TENSORQ}(\mathbb{F}, n) := \big\{(a, b) \in \mathbb{F}^n \times \mathbb{F}^{n^2} \mid b = \mathrm{flat}(a \otimes a)\big\}.$$

Definition 3.5.2. Given $(a, b) \in \mathbb{F}^n \times \mathbb{F}^{n^2}$, the LPCP verifier $V^{\langle \pi, \cdot \rangle}_{\mathrm{TENSORQ}(\mathbb{F},n)}(a, b)$ uses a proof $\pi = \alpha \parallel \beta \in \mathbb{F}^{n+n^2}$, intended to be the linear encoding of the public input (a, b). Since the proof may be dishonest, the verifier checks consistency with (a, b) on the sampled queries before checking the tensor relation:

1. Sample $s, t \leftarrow \mathbb{F}^n$.
2. Query the α-part of π at s and at t, obtaining y_s and y_t.
3. Query the β-part of π at $\mathrm{flat}(s \otimes t)$, obtaining $y_{s,t}$.
4. Accept if and only if

$$y_s = \langle a, s\rangle, \qquad y_t = \langle a, t\rangle, \qquad y_{s,t} = \langle b, \mathrm{flat}(s \otimes t)\rangle, \qquad y_{s,t} = y_s y_t.$$

Definition 3.5.3. In settings where a and b are supplied only by the proof oracle, the tensor self-check uses the same three queries but has no public input to compare against. Given oracle access to $\pi = \alpha \parallel \beta \in \mathbb{F}^{n+n^2}$, sample $s, t \leftarrow \mathbb{F}^n$, query

$$y_s = \langle \alpha, s\rangle, \qquad y_t = \langle \alpha, t\rangle, \qquad y_{s,t} = \langle \beta, \mathrm{flat}(s \otimes t)\rangle,$$

and accept if and only if $y_{s,t} = y_s y_t$. This is the tensor component used inside the QESAT verifier.

Lemma 3.5.4. *For every $a, s, t \in \mathbb{F}^n$,*

$$\langle \mathrm{flat}(a \otimes a), \mathrm{flat}(s \otimes t)\rangle = \langle a, s\rangle\langle a, t\rangle.$$

Proof.

$$\langle \mathrm{flat}(a \otimes a), \mathrm{flat}(s \otimes t)\rangle = \sum_{i,j} a_i a_j s_i t_j = \left(\sum_i a_i s_i\right)\left(\sum_j a_j t_j\right) = \langle a, s\rangle\langle a, t\rangle.$$

□

Lemma 3.5.5. *Let $\mathbb{F}$ be a finite field. If $b \neq \mathrm{flat}(a \otimes a)$, then for every proof $\pi \in \mathbb{F}^{n+n^2}$,*

$$\Pr_{s,t}\Big[V^{\langle \pi, \cdot \rangle}_{\mathrm{TENSORQ}(\mathbb{F},n)}(a, b) = 1\Big] \leq \frac{2|\mathbb{F}| - 1}{|\mathbb{F}|^2}.$$

Proof. Suppose $b \neq \mathrm{flat}(a \otimes a)$, i.e., there exists $i^*, j^* \in [n]$ such that $b_{i^*j^*} \neq a_{i^*} \cdot a_{j^*}$. For each $i \in [n]$ define the linear polynomial $p_i(t) := \sum_{j\in[n]} (b_{ij} - a_i a_j)\, t_j$. For fixed s, t, if the verifier accepts, then its consistency checks force $y_s = \langle a, s\rangle$, $y_t = \langle a, t\rangle$, and $y_{s,t} = \langle b, \mathrm{flat}(s \otimes t)\rangle$. Thus acceptance can only occur when

$$\langle b, \mathrm{flat}(s \otimes t)\rangle - \langle a, s\rangle\langle a, t\rangle = \sum_{i\in[n]} \sum_{j\in[n]} (b_{ij} - a_i a_j)\, s_i t_j = \sum_{i\in[n]} p_i(t)\, s_i.$$

Immediate application of Polynomial Identity Testing yields a lower bound of $1 - \frac{2}{|\mathbb{F}|}$ but we can do better. Since $b_{i^*j^*} \neq a_{i^*} a_{j^*}$, the polynomial p_{i^*} is nonzero, so by the Schwartz–Zippel lemma applied to t,

$$\Pr_t[p_{i^*}(t) \neq 0] \geq 1 - \tfrac{1}{|\mathbb{F}|}.$$

Conditioned on $p_{i^*}(t) \neq 0$, the expression $\sum_i p_i(t)\, s_i$ is a nonzero linear polynomial in s, so by the Schwartz–Zippel lemma applied to s,

$$\Pr_{s\leftarrow\mathbb{F}^n}\left[\sum_i p_i(t)\, s_i \neq 0 \mid p_{i^*}(t) \neq 0\right] \geq 1 - \tfrac{1}{|\mathbb{F}|}.$$

Hence

$$\Pr_{s,t}\left[V^{\langle\pi,\cdot\rangle}_{\mathrm{TENSORQ}(\mathbb{F},n)}(a, b) = 0\right] \geq \left(1 - \tfrac{1}{|\mathbb{F}|}\right)^2,$$

and therefore

$$\Pr_{s,t}\left[V^{\langle\pi,\cdot\rangle}_{\mathrm{TENSORQ}(\mathbb{F},n)}(a, b) = 1\right] \leq 1 - \left(1 - \tfrac{1}{|\mathbb{F}|}\right)^2 = \frac{2|\mathbb{F}| - 1}{|\mathbb{F}|^2}.$$

□

Lemma 3.5.6. *Let $\mathbb{F}$ be a finite field and let $\pi = \alpha \parallel \beta \in \mathbb{F}^{n+n^2}$. If $(\alpha, \beta) \notin \mathrm{TENSORQ}(\mathbb{F}, n)$, then the tensor self-check accepts with probability at most*

$$\frac{2|\mathbb{F}| - 1}{|\mathbb{F}|^2}.$$

Proof. Apply 3.5.5 to the public pair (α, β). The self-check is exactly the multiplicativity part of that verifier, with (α, β) read from the proof oracle itself. □

Theorem 3.5.7. *For every finite field $\mathbb{F}$ and $n \in \mathbb{N}$,*

$$\mathrm{TENSORQ}(\mathbb{F}, n) \in \mathrm{LPCP}\left[\varepsilon_c = 0, \varepsilon_s = \frac{2|\mathbb{F}| - 1}{|\mathbb{F}|^2}, \Sigma = \mathbb{F}, \ell = n + n^2, q = 3, r = 2n\right].$$

Proof. Use the verifier of 3.5.2. If $(a, b) \in \mathrm{TENSORQ}(\mathbb{F}, n)$, the honest proof $\pi = (a, b)$ passes the three input-consistency checks identically, and the multiplicativity check is 3.5.4. Soundness is 3.5.5. The verifier makes three proof queries and uses two random vectors in $\mathbb{F}^n$, which gives the displayed query and randomness bounds. □

Corollary 3.5.8.

$$\mathrm{TENSORQ}(\mathbb{F}_2, n) \in \mathrm{LPCP}[\varepsilon_c = 0, \varepsilon_s = 3/4, \Sigma = \mathbb{F}_2, \ell = n + n^2, q = 3, r = 2n].$$

Proof. Specialize 3.5.7 to $\mathbb{F}_2$, where $|\mathbb{F}_2| = 2$. □

3.6 Linear PCP to PCP

Lemma 3.6.1 (Chernoff bound)**.** *Let $X_1, \dots, X_N$ be independent random variables over $\{0,1\}$, and let $\mu = \mathbb{E}\left[\sum_{i=1}^N X_i\right]$. For any $\Delta > 0$ one has*

$$\Pr\left[\sum_{i=1}^{N} X_i \geq \mu + \Delta\right] \leq \exp(-2\Delta^2/N)$$

Proof. Write $p_i = \mathbb{E}[X_i]$, so that $\mu = \sum_{i=1}^N p_i$. We first prove the elementary exponential-moment estimate

$$\mathbb{E}[\exp(\lambda(X_i - p_i))] \leq \exp(\lambda^2/8)$$

for every $\lambda > 0$ and every i. Since X_i is $\{0,1\}$-valued,

$$\mathbb{E}[\exp(\lambda(X_i - p_i))] = (1 - p_i)\exp(-\lambda p_i) + p_i \exp(\lambda(1 - p_i)).$$

Equivalently, it is enough to show

$$\log(1 - p_i + p_i \exp(\lambda)) - p_i\lambda \leq \lambda^2/8.$$

For $p \in [0,1]$, set $f(\lambda) = \log(1 - p + p\exp(\lambda)) - p\lambda$. Then $f(0) = f'(0) = 0$, and

$$f''(\lambda) = \frac{p\exp(\lambda)}{1 - p + p\exp(\lambda)}\left(1 - \frac{p\exp(\lambda)}{1 - p + p\exp(\lambda)}\right) \leq \frac{1}{4}.$$

Hence, for $\lambda > 0$,

$$f(\lambda) = \int_0^\lambda \int_0^u f''(v)\,dv\,du \leq \int_0^\lambda \int_0^u \frac{1}{4}\,dv\,du = \lambda^2/8.$$

Let $S = \sum_{i=1}^N X_i$. By independence and the estimate above,

$$\mathbb{E}[\exp(\lambda(S - \mu))] = \prod_{i=1}^{N} \mathbb{E}[\exp(\lambda(X_i - p_i))] \leq \exp(N\lambda^2/8).$$

Markov's inequality gives, for every $\lambda > 0$,

$$\Pr[S \geq \mu + \Delta] = \Pr[\exp(\lambda(S - \mu)) \geq \exp(\lambda\Delta)] \leq \exp(-\lambda\Delta)\mathbb{E}[\exp(\lambda(S - \mu))]$$

Using the moment bound above, this is at most

$$\exp(-\lambda\Delta + N\lambda^2/8).$$

Taking $\lambda = 4\Delta/N$ gives

$$\Pr[S \geq \mu + \Delta] \leq \exp(-2\Delta^2/N).$$

□

Definition 3.6.2. For an LPCP verifier making at most q linear queries, set

$$t := 8(\lfloor \log_2(100q) \rfloor + 1).$$

This is the concrete logarithmic shift count used in Lean. The converted PCP verifier uses t independent random shifts to decode each linear query by plurality.

Definition 3.6.3 (Encoded proof)**.** For a linear LPCP proof $\pi \in \mathbb{F}_2^\ell$, its PCP encoding $\widehat{\pi}_{\text{lin}} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ is the truth table

$$\widehat{\pi}_{\text{lin}}(a) := \langle \pi, a \rangle, \qquad a \in \mathbb{F}_2^\ell.$$

Definition 3.6.4. Given a PCP proof $\widehat{\pi} : \mathbb{F}_2^\ell \to \mathbb{F}_2$, shifts $r_1, \ldots, r_t \in \mathbb{F}_2^\ell$, and a linear query $a \in \mathbb{F}_2^\ell$, define the locally corrected answer

$$\mathsf{Corr}_{\widehat{\pi}}(a; r_1, \ldots, r_t) := \text{plurality}_{i \in [t]}(\widehat{\pi}(a + r_i) - \widehat{\pi}(r_i)).$$

Definition 3.6.5. Let V be an LPCP verifier over $\mathbb{F}_2$. The converted PCP verifier $\overline{V}$, with oracle access to $\widehat{\pi} : \mathbb{F}_2^\ell \to \mathbb{F}_2$, proceeds as follows on input x:

1. If $V^{\widehat{\pi}}_{\text{BLR}(\mathbb{F}_2, n)} = 0$, reject. Otherwise, proceed to the next step.
2. Sample t shifts $r_1, \ldots, r_t \in \mathbb{F}_2^\ell$, with t as in 3.6.2.
3. Simulate $V(x)$. Whenever V asks a linear query $a \in \mathbb{F}_2^\ell$, answer it by $\mathsf{Corr}_{\widehat{\pi}}(a; r_1, \ldots, r_t)$.

Lemma 3.6.6 (Completeness)**.** *If* $\Pr\left[V^{\langle \pi, \cdot \rangle}(x) = 1\right] \geq 1 - \varepsilon_c$*, then* $\Pr\left[\overline{V}^{\widehat{\pi}_{\text{lin}}}(x) = 1\right] \geq 1 - \varepsilon_c$.

Proof. The encoded proof is linear, so the BLR test accepts with probability 1 by 3.3.2. For every query a and every shift r_i,

$$\widehat{\pi}_{\text{lin}}(a + r_i) - \widehat{\pi}_{\text{lin}}(r_i) = \langle \pi, a \rangle.$$

Thus every plurality answer equals the corresponding LPCP oracle answer, and the simulation has exactly the same acceptance probability as V. ☐

Lemma 3.6.7 (Soundness, far case)**.** *If* $\delta(\widehat{\pi}, \mathcal{LIN}) \geq 1/8$*, then*

$$\Pr[\overline{V}^{\widehat{\pi}}(x) = 1] \leq 7/8.$$

Proof. By 3.3.3, the BLR precheck accepts with probability at most $1 - \delta(\widehat{\pi}, \mathcal{LIN}) \leq 7/8$. Since $\overline{V}$ accepts only when this test accepts, its acceptance probability is at most $7/8$. ☐

Lemma 3.6.8. *If* $\delta(\widehat{\pi}, f) < 1/8$ *for some linear function* f *of the form* $u \mapsto \langle \pi, u \rangle$*, then*

$$\Pr_{r_1, \ldots, r_t}[\mathsf{Corr}_{\widehat{\pi}}(a; r_1, \ldots, r_t) \neq \langle \pi, a \rangle] \leq \exp(-t/8), \quad \forall a \in \mathbb{F}_2^\ell.$$

Proof. For one shift r, the corrected sample can be wrong only if $\widehat{\pi}(a + r)$ or $\widehat{\pi}(r)$ disagrees f. A union bound gives error probability at most $2 \cdot (1/8) = 1/4$. If X_i is the indicator of failure for shift r_i, then $\mathbb{E}[\sum_i X_i] \leq t/4$. The plurality answer is wrong only if $\sum_i X_i \geq t/2$, so 3.6.1 with $\Delta = t/4$ gives $\exp(-t/8)$. ☐

Lemma 3.6.9. *Assume every possible linear query* a *is locally corrected incorrectly with probability at most* η*. If the simulated LPCP computation makes at most* q *proof queries, then its acceptance probability under the corrected PCP oracle is at most the acceptance probability under the nearby linear proof plus* $q\eta$*.*

Proof. Expose the LPCP computation one oracle query at a time. At a proof query, the two simulations can diverge only when the local correction for the queried linear form is wrong; by assumption this has probability at most η. Inducting over the query-bound derivation gives an additive loss of at most $q\eta$. ☐

Lemma 3.6.10 (Soundness, near case)**.** *If $x \notin L$, V has LPCP soundness ε_s, and $\delta(\hat{\pi}, f) < 1/8$ for some linear function f of the form $u \mapsto \langle \pi, u\rangle$, then*

$$\Pr[\overline{V}^{\hat{\pi}}(x) = 1] \leq \varepsilon_s + \frac{1}{100}.$$

Proof. By 3.6.8, every possible linear query has local correction error at most $\eta = \exp(-t/8)$. The hybrid estimate 3.6.9 compares the corrected simulation with the simulation using the nearby linear proof and loses at most $q\eta$. The choice of t in 3.6.2 gives $q\eta \leq 1/100$. The nearby linear proof is a valid LPCP proof, so its acceptance probability is at most ε_s because $x \notin L$. ☐

Lemma 3.6.11 (Soundness)**.** *For every $x \notin L$ and every PCP proof $\hat{\pi}$,*

$$\Pr[\overline{V}^{\hat{\pi}}(x) = 1] \leq \max\{7/8, \varepsilon_s + 1/100\}.$$

Proof. If $\delta(\hat{\pi}, \mathcal{LIN}) \geq 1/8$, apply 3.6.7. Otherwise, apply 3.6.10. ☐

Lemma 3.6.12 (LPCPs admit exponential-size PCPs)**.**

$$\mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r] \subseteq$$
$$\mathrm{PCP}\left[\varepsilon_c,\ \varepsilon_s' = \max\{7/8, \varepsilon_s + 1/100\}, \Sigma' = \mathbb{F}_2,\ \ell' = 2^\ell,\ q' = 3 + 2tq,\ r' = r + (2+t)\ell\right].$$

Proof. Let $L \in \mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r]$ and let V be the LPCP verifier for L. Use the converted PCP verifier $\overline{V}$ defined in 3.6.5. Completeness is 3.6.6, and soundness is 3.6.11. The BLR test makes three proof queries. Each simulated LPCP query uses two PCP queries for each of the t shifts, so the query bound is $3 + 2tq$. The randomness consists of the original r random bits, 2ℓ bits for the BLR test, and $t\ell$ bits for the shifts. ☐

3.7 PCP for QESAT

Let $x = (p_1, \dots, p_m)$ be a QESAT$(\mathbb{F}_2, n)$ instance. We use the following linearization. For each quadratic polynomial p_i, form a row $M_i \in \mathbb{F}_2^{n+n^2}$ by replacing every linear monomial X_j by the variable a_j and every quadratic monomial $X_j X_k$ by the tensor variable b_{jk}. Let

$$M_x := \begin{bmatrix} M_1 \\ \vdots \\ M_m \end{bmatrix} \in \mathbb{F}_2^{m \times (n+n^2)} \quad \text{and} \quad c_x := \begin{bmatrix} -\mathrm{const}(p_1) \\ \vdots \\ -\mathrm{const}(p_m) \end{bmatrix} \in \mathbb{F}_2^m,$$

where $\mathrm{const}(p_i)$ is the constant term of p_i. Then $M_x(a \parallel b) = c_x$ expresses the original equations after replacing $a_j a_k$ by b_{jk}.

Definition 3.7.1. Let $x = (p_1, \dots, p_m) \in R_2^m$ be a QESAT instance as defined in 3.0.1. Let $V^{\pi}_{\mathrm{QESAT}(\mathbb{F}_2,n)}(x)$ be the LPCP verifier with oracle access to $\pi = a \parallel b \in \mathbb{F}_2^{n+n^2}$, defined as follows:

1. Sample $r \leftarrow \mathbb{F}_2^m$ and $s, t \leftarrow \mathbb{F}_2^n$.
2. Construct $(M_x, c_x) \in \mathbb{F}_2^{m \times (n+n^2)} \times \mathbb{F}_2^m$ by the linearization above.
3. Check that $V^{\langle \pi, \cdot \rangle}_{\mathrm{LINEQ}(\mathbb{F}_2, m, n+n^2)}(M_x, c_x) = 1$.

4. Query π at $(s, 0)$, $(t, 0)$, and $(0, \mathrm{flat}(s \otimes t))$ and check that
$$\langle b, \mathrm{flat}(s \otimes t)\rangle = \langle a, s\rangle\langle a, t\rangle.$$

Equivalently, this last step is the tensor self-check of 3.5.3 specialized to $\mathbb{F}_2$.

Lemma 3.7.2 (Completeness)**.** *If $x \in \mathrm{QESAT}(\mathbb{F}_2, n)$, then exists a proof $\pi \in \mathbb{F}_2^{n+n^2}$ such that*
$$\Pr\left[V^{\pi}_{\mathrm{QESAT}(\mathbb{F}_2,n)}(x) = 1\right] = 1.$$

Proof. Let $a \in \mathbb{F}_2^n$ satisfy all equations and set $b = \mathrm{flat}(a \otimes a)$. By construction of the linearized system, $M_x(a \parallel b) = c_x$, so the LINEQ check accepts for every r. The tensor self-check accepts for every s, t by 3.5.4. □

Lemma 3.7.3 (Soundness)**.** *If $x \notin \mathrm{QESAT}(\mathbb{F}_2, n)$, then for every proof $\tilde{\pi} \in \mathbb{F}_2^{n+n^2}$,*
$$\Pr\left[V^{\tilde{\pi}}_{\mathrm{QESAT}(\mathbb{F}_2,n)}(x) = 1\right] \leq 3/4.$$

Proof. If some p_i has total degree > 2, the verifier rejects immediately, so the acceptance probability is 0. Assume now that every p_i is quadratic and write $\pi = a \parallel b$. If $b \neq \mathrm{flat}(a \otimes a)$, then the tensor self-check accepts with probability at most 3/4 by 3.5.6 specialized to $\mathbb{F}_2$, and the full verifier accepts only if that check accepts.

Otherwise $b = \mathrm{flat}(a \otimes a)$. Since $x \notin \mathrm{QESAT}(\mathbb{F}_2, n)$, the assignment a does not satisfy all equations. By construction of the linearized system, $M_x(a \parallel b) \neq c_x$. The LINEQ check therefore accepts with probability at most $1/|\mathbb{F}_2| = 1/2$ by 3.4.3. Again the full verifier accepts only if this check accepts, so the acceptance probability is at most $1/2 \leq 3/4$. □

Theorem 3.7.4.
$$\mathrm{QESAT}(\mathbb{F}_2, n) \in \mathrm{LPCP}[\varepsilon_c = 0, \varepsilon_s = 3/4, \Sigma = \mathbb{F}_2, \ell = n + n^2, q = 4, r = |x| + 2n].$$

Proof. Use the verifier 3.7.1. It makes one query for the LINEQ check and three queries for the tensor self-check, for a total of four. Its randomness is the LINEQ randomness plus two vectors in $\mathbb{F}_2^n$. Completeness and soundness are 3.7.2 and 3.7.3. □

Theorem 3.7.5.
$$\mathrm{QESAT}(\mathbb{F}_2, n) \in \mathrm{PCP}[\varepsilon_c = 0, \varepsilon_s = 7/8, \Sigma = \mathbb{F}_2, \ell = 2^{|x|}, q = O(1), r = |x|^{O(1)}].$$

Proof. Apply 3.6.12 to 3.7.4. Since the LPCP soundness is 3/4, the converted soundness is
$$\max\{7/8,\ 3/4 + 1/100\} = 7/8.$$
The query bound remains constant because $q = 4$, and the randomness is polynomial in the input size. The Lean formalization pads the proof oracle from length 2^{n+n^2} to length $2^{|x|}$. □

Lemma 3.7.6.
$$\mathrm{PCP}[0, 7/8, \mathbb{F}_2, 2^{|x|}, O(1), |x|^{O(1)}] \subseteq \mathrm{PCP}[0, 1/2, \mathbb{F}_2, 2^{|x|}, O(1), |x|^{O(1)}].$$

Proof. Invoke 3.7.5 to obtain the 7/8-sound verifier. Run it independently six times and accept only if all six runs accept. Completeness error remains 0. For $x \notin L$, the acceptance probability is at most $(7/8)^6 \leq 1/2$. Query and randomness bounds are multiplied by the constant 6. □

Theorem 3.7.7.
$$\mathrm{QESAT}(\mathbb{F}_2, n) \in \mathrm{PCP}[\varepsilon_c = 0, \varepsilon_s = 1/2, \Sigma = \mathbb{F}_2, \ell = 2^{|x|}, q = O(1), r = |x|^{O(1)}].$$

Proof. Combine 3.7.5 with the constant repetition step 3.7.6. □

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